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September 1, 2026

Wellcome Leap

Resistance Networks Program

Dear Members of the Resistance Networks Review Committee:

I am an Associate Research Professor at the Centro de Ciencias Matemáticas, Universidad Nacional Autónoma de México, where my research focuses on mathematical epidemiology, infectious disease modeling, Bayesian inference, and uncertainty quantification. My work has developed mathematical and statistical methods for inferring transmission dynamics from epidemiological data, including the estimation of reproduction numbers and uncertainty in partially observed epidemic processes.

The proposal *Universal Properties of Plasmid Evolution and Robust Risk Forecasting for Preemptive pARG Intervention* seeks to determine whether the spread of antibiotic-resistance plasmids can be quantified and forecast using concepts from epidemic theory.

The Bayesian framework developed in Trejo et al. (2024) provides a strong methodological foundation for this component of the proposed work. Our approach separates two fundamental components of epidemic transmission. The first is a time-varying reproductive rate describing the population-level transmission process. The second is an infection profile describing how infectiousness varies over time following infection within the host. We combine these components in a renewal-type compartmental model with a negative-binomial likelihood for incidence observations. The reproductive rate is represented nonparametrically using splines with a Bayesian smoothness prior, and Markov chain Monte Carlo inference provides posterior estimates of the reproductive rate, the effective reproduction number, and the predictive distribution of incidence.

This framework is directly relevant to plasmid transmission. For a plasmid epidemic, the incidence data would consist of newly observed plasmid acquisitions. The infection profile would be replaced by a kernel describing the expected transfer-generating capacity of a newly plasmid-bearing cell or lineage as a function of time or generations since acquisition.

This kernel could incorporate the onset and duration of transfer competence, plasmid persistence, segregational loss, and, where appropriate, clonal expansion. The time-varying reproductive rate would describe changes in the external conditions governing transfer, including antibiotic treatment, donor-recipient density, community composition, environmental conditions, and changes in plasmid or host genotype.



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Importantly, this framework provides a natural connection between controlled experiments and longitudinal surveillance. Controlled experiments can be used to estimate the plasmid-specific transfer and persistence kernel. Longitudinal observations can then be used to infer the reproductive rate with associated uncertainty. The negative-binomial observation model is appropriate for overdispersed event counts, although an additional observation layer will be needed when plasmid acquisitions are incompletely detected or when available measurements report prevalence rather than incidence.

Our previous work also provides a useful basis for evaluating the robustness of reproduction-number estimates to uncertainty in the underlying transmission profile. In our analysis of COVID-19 data, estimates of the effective reproduction number were less sensitive to plausible changes in the infection profile than estimates of the instantaneous reproductive rate. Whether an analogous robustness property holds for plasmid transmission is an important empirical question that can be addressed through the proposed experiments and sensitivity analyses.

The nonparametric Bayesian framework is particularly useful for surveillance, retrospective identification of transmission waves, and validation of mechanistic models. It does not, without additional assumptions, determine the future trajectory of the reproductive rate. Coupling it to the proposed mechanistic models of plasmid-host evolution is therefore appropriate: the Bayesian model can provide a data-driven estimate and uncertainty benchmark, while the mechanistic model supplies intervention-specific forecasts.

I strongly support the proposed work and believe that this integration of epidemiological inference with plasmid and host evolutionary dynamics has the potential to establish a quantitative framework for identifying when resistance transmission has epidemic-like potential. If funded, I would contribute to the adaptation of our Bayesian modeling framework, including likelihood formulation, transmission kernel, inference procedures, uncertainty quantification, and model-validation strategies, to plasmid transmission data.

Sincerely,

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